

DUMONT D'URVILLE STATION, Antarctica

WMO: 896420

Lat: 66.663S

Lon: 140.001E

Elev: 43

StdP: 100.81

Time Zone: 10.00 (E10)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-28.3	-26.8	-34.6	0.1	-26.6	-33.2	0.2	-25.1	30.6	-13.0	28.1	-13.5	6.3	140	1.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.7	3.1	-0.4	2.1	-0.9	1.3	-1.4	0.2	2.0	-0.4	1.4	-0.9	0.9	6.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
-1.1	3.5	0.0	-2.0	3.2	-0.3	-3.0	3.0	-0.5	10.2	2.2	9.1	1.4	8.1	1.0	2.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.2	24.0	21.1	-30.0	4.8	0.9	1.8	-30.7	6.1	-31.2	7.1	-31.7	8.1	-32.4	9.3		
(5)			-29.9	1.2	1.4	0.8	-30.9	1.8	-31.8	2.3	-32.6	2.7	-33.6	3.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-11.2	-1.0	-4.5	-9.6	-14.1	-16.2	-16.0	-17.2	-17.1	-15.4	-13.4	-7.0	-1.9	(6)
(7)		DBStd	6.99	1.88	2.50	3.33	3.75	4.89	4.67	4.98	5.21	4.55	3.48	3.13	1.87	(7)
(8)		HDD10.0	7723	342	407	609	722	813	779	845	842	761	726	509	368	(8)
(9)		HDD18.3	10765	601	640	867	972	1072	1029	1103	1100	1011	984	759	627	(9)
(10)		CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	8.9	7.8	10.0	10.0	9.4	8.9	9.2	8.5	8.7	10.1	8.5	8.2	8.0	(14)
(15)	Precipitation	PrecAvg	508	59	60	53	37	33	33	44	42	39	28	31	50	(15)
(16)		PrecMax	688	120	125	87	78	56	77	81	69	65	50	49	108	(16)
(17)		PrecMin	402	22	21	23	20	12	12	8	22	12	7	14	12	(17)
(18)		PrecStd	71	27	24	18	13	11	14	17	13	14	12	10	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.5	2.2	-0.7	-2.8	-2.9	-4.7	-4.3	-2.4	-2.7	-3.6	2.0	4.0	(19)
(20)			MCWB	0.8	-1.1	-1.9	-3.5	-3.2	-5.9	-5.7	-3.2	-3.5	-5.4	-1.2	0.0	(20)
(21)		2%	DB	3.6	1.0	-2.0	-5.9	-5.6	-6.5	-6.5	-5.1	-5.7	-5.8	0.4	2.6	(21)
(22)			MCWB	0.0	-1.7	-3.4	-7.2	-7.0	-7.7	-8.0	-5.9	-7.1	-7.5	-2.6	-0.7	(22)
(23)		5%	DB	2.6	0.2	-3.3	-7.8	-7.9	-8.0	-8.6	-7.8	-7.2	-7.1	-0.9	1.7	(23)
(24)			MCWB	-0.6	-2.1	-4.8	-9.1	-9.1	-9.5	-9.9	-8.6	-8.6	-8.7	-3.7	-1.3	(24)
(25)		10%	DB	1.8	-0.6	-4.8	-9.1	-9.8	-9.3	-10.3	-9.9	-8.9	-8.4	-2.1	1.0	(25)
(26)			MCWB	-1.0	-2.8	-6.6	-10.4	-10.9	-10.5	-11.3	-10.6	-10.0	-9.8	-4.6	-1.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	-0.3	-1.3	-3.4	-2.9	-5.8	-4.9	-3.1	-3.6	-5.0	-0.4	0.6	(27)
(28)			MCDB	4.0	0.8	-0.9	-2.9	-2.9	-5.0	-4.6	-2.8	-2.9	-3.4	1.3	2.8	(28)
(29)		2%	WB	0.6	-1.1	-3.2	-7.2	-7.2	-7.4	-7.9	-5.8	-6.9	-7.2	-2.2	-0.2	(29)
(30)			MCDB	2.7	0.4	-2.2	-6.0	-5.9	-6.7	-6.7	-5.4	-6.0	-6.0	0.0	1.7	(30)
(31)		5%	WB	0.0	-1.8	-4.7	-8.9	-9.0	-9.2	-9.9	-8.6	-8.5	-8.5	-3.5	-0.8	(31)
(32)			MCDB	1.8	-0.1	-3.4	-7.9	-8.0	-8.2	-8.9	-7.9	-7.4	-7.3	-1.3	1.1	(32)
(33)		10%	WB	-0.5	-2.6	-6.5	-10.2	-10.8	-10.6	-11.2	-10.7	-9.9	-9.8	-4.5	-1.3	(33)
(34)			MCDB	1.3	-0.7	-5.0	-9.3	-9.9	-9.4	-10.2	-9.9	-8.8	-8.5	-2.2	0.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.7	4.2	4.3	4.0	4.2	4.3	4.8	4.7	5.2	5.4	4.2	(35)	
(36)			MCDBR	4.5	4.4	4.1	4.6	4.3	4.6	5.5	5.4	5.1	5.6	5.6	4.6	(36)
(37)			MCWBR	3.3	3.6	3.6	4.2	3.9	4.2	5.1	5.0	4.6	5.0	4.7	3.6	(37)
(38)		5% WB	MCDBR	3.6	3.7	3.7	4.7	4.2	4.4	5.2	5.1	4.9	5.3	5.2	4.1	(38)
(39)			MCWBR	2.9	3.3	3.6	4.4	3.9	4.2	5.0	5.0	4.5	4.9	4.6	3.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.261	0.246	0.222	0.178	0.120	N/A	0.121	0.182	0.210	0.221	0.234	0.260	(40)	
(41)		taud	2.325	2.353	2.328	2.186	1.917	N/A	1.907	2.152	2.251	2.272	2.284	2.283	(41)	
(42)		Ebn at Noon	1000	963	890	723	358	0	321	693	886	986	1028	1015	(42)	
(43)		Edh at Noon	109	92	71	42	12	0	11	43	74	98	112	118	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.55	5.31	2.96	1.08	0.17	0.00	0.06	0.59	2.23	4.96	7.75	8.70	(44)	
(45)		RadStd	0.39	0.32	0.16	0.04	0.01	0.00	0.00	0.04	0.10	0.19	0.22	0.39	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page